

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

Degree, Semester & Branch	: II Semester B.E.EEE
Course Code & Title	: EE8251 – Circuit Theory
Name of the Faculty member	: Dr. S. Kannan
Name of the Topic	: Current flow in series and parallel circuits
Name of the Innovative Practice	: Simulation
Date & Duration	: -
ICT Tool Used	: EveryCircuit Simulation Software

Description:

- Utilized the EveryCircuit simulation software to visualize the current flow in the DC circuits and for better understanding of the series and parallel resistor combination.
- Taught both Android & Chrome OS based EveryCircuit simulation software, so that the students can learn the analysis of circuits on the go.

Goals (Learning Outcomes):

- Students will be able to understand the current flow in series and parallel circuits.

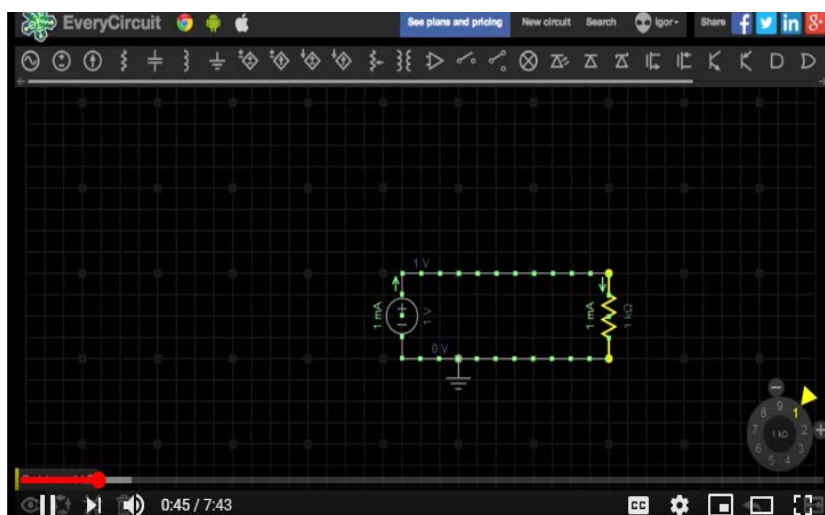
Use of appropriate methods:

- Using simulation software's only students can able to visualize the current flow direction in the circuits.

Effective presentation

Details of the Implementation:

After teaching the concepts of current flow in series and parallel circuits, the students were exposed to do the simulation.



Total No. of Students : 26

Time allotted for the activity : 50 minutes

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- At the end of the class (last 10 minutes) I asked the students to summarize how to simulate a circuit using everycircuit. Some of the steps they missed during the summary and finally I insisted the points why they missed.
- The success of activity was evaluated by asking the same question in Internal Assessment Test and around 80% of the students answered correctly.

Reflective Critique:

Benefits:

- Students were exposed to new simulation software which is not in their curriculum.

Challenges:

- Time utilization for conducting activity.
- Biology background students struggle a lot during activity.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	3	-	-	-	-	-	2	3

References:

1. "Engineering Circuit Analysis" by William H. Hayt and Jack E. Kemmerly.
2. Sonaversity animation CD for Circuit Theory.